Cell-Type-Aware Prediction of Enhancer Activity Using Convolutional Neural Networks

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1 Abstract

Enhancer activity is highly dependent on cellular context, making its prediction from DNA sequence a challenging task. This study evaluated whether incorporating cell-type information improves enhancer activity prediction using convolutional neural networks (CNNs). The study compared four CNN architectures (sequence-only, embedding-based, attention-based, and combined) using MPRA data from three human cell types. Models incorporating cell-type embeddings improved aggregate performance on the combined test set and outperformed the sequence-only and attention-based approaches overall, while attention mechanisms provided limited benefit. These results suggest that cell-type information contributes to more robust enhancer activity prediction and highlight the importance of cellular context in regulatory genomics.

2 Introduction

Gene expression is controlled by a complex network of regulatory elements that determine when and where genes are activated. Among these elements, enhancers play a crucial role by increasing transcriptional activity through interactions with transcription factors and other regulatory proteins. Unlike protein-coding sequences, enhancer function depends not only on its DNA sequence but also on the cellular environment in which the sequence is interpreted [1].

Recent advances in massively parallel reporter assays (MPRAs) have enabled large-scale functional characterization of regulatory DNA elements. By testing thousands of candidate enhancers simultaneously, MPRAs provide quantitative measurements of regulatory activity and have become an important tool for studying gene regulation [2]. Large datasets generated by these experiments have also created opportunities for applying machine learning methods to predict enhancer activity directly from DNA sequence.

Deep learning approaches have shown noticeable success in genomic sequence analysis [3]. One of the most widely used architectures are convolutional neural networks (CNNs) which are able to automatically learn sequence motifs and patterns associated with regulatory functions. However, many predictive models focus primarily on sequence information, and do not explicitly incorporate cellular context, despite the well-established cell-type specificity of enhancer activity.

The dataset used in this study comes from a large-scale ENCODE project [4], specifically from Phase 4 MPRA experiment that measured regulatory activity across multiple human cell types, including HepG2, K562, and WTC11 cells [5]. The availability of activity measurements for the same regulatory elements in different cellular contexts provided an opportunity to investigate whether incorporating cell-type information can improve predictive performance.

The primary hypothesis of this study is that cell-type information improves enhancer activity prediction beyond what can be achieved using DNA sequences only. To test this hypothesis, four CNN-based architectures were developed and compared. Model performance was further evaluated using multiple regression metrics together with separate analysis of predictive accuracy for each cell type.

3 Materials and Methods

3.1 Dataset

The dataset used in this study was derived from a large-scale ENCODE Phase 4 Massively Parallel Reporter Assay (MPRA) experiment. The analyzed joint MPRA library consisted primarily of potential enhancers, together with a smaller number of promoters and experimental control sequences, assayed across three human cell types: HepG2, K562, and WTC11. For each element, activity was quantified using the $\log_2(\mathrm{RNA}/\mathrm{DNA})$ ratio, providing a continuous measure of enhancer activity.

DNA sequences were obtained from the shared MPRA element library, while activity measurements for each cell type were provided separately. The same regulatory elements were assayed across multiple cellular contexts, making the dataset suitable for investigating the contribution of cell-type information to enhancer activity prediction.

3.2 Data Preprocessing

DNA sequences were matched with their corresponding activity measurements using unique enhancer identifiers. Activity values were aggregated by averaging replicate measurements for each element–cell-type pair.

Sequences were represented using one-hot encoding, where each nucleotide was converted into a four-dimensional binary vector. Specifically, adenine (A), thymine (T), guanine (G) and cytosine (C) were encoded as orthogonal vectors, while unknown nucleotides (N) were represented by zeros.

Cell types were encoded as integer labels: HepG2 = 0, K562 = 1, and WTC11 = 2.

To prevent data leakage, the dataset was split according to enhancer identifiers rather than individual observations. This ensured that the same enhancer sequence could not appear simultaneously in both training and testing subsets. The data were divided into training (70%), validation (15%) and test (15%) sets.

3.3 Model Architectures

Four convolutional neural network architectures were evaluated.

Baseline CNN

The baseline model used only DNA sequence information. Input sequences were processed by three one-dimensional convolutional layers with ReLU activation functions and batch normalization. Global average pooling was applied to obtain a fixed-length sequence representation, followed by a fully connected regression head.

Embedding CNN

The embedding model extended the baseline architecture by incorporating a learned cell-type embedding. A trainable embedding vector was assigned to each cell type and concatenated with the sequence representation before the final regression layers.

Attention CNN

This model replaced global average pooling with an attention-based pooling mechanism. Attention weights were learned across sequence positions, allowing the model to assign different importance to individual positions when aggregating the final convolutional feature map.

Full CNN

The final architecture combined both approaches by incorporating cell-type embeddings together with attention-based pooling.

3.4 Training Procedure

All models were implemented in PyTorch and trained as regression models. Mean squared error (MSE) was used as the optimization objective.

Model parameters were optimized using the Adam optimizer with a learning rate of 10^{-4} and weight decay of 10^{-4}. Training was performed for 30 epochs using a batch size of 64. Validation loss was monitored throughout training and subsequently used to compare model variants.

3.5 Evaluation Metrics

Model performance was evaluated on the held-out test set using multiple regression metrics.

Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Squared Error (MSE) were used to quantify prediction error. Pearson and Spearman correlation coefficients were calculated to assess linear and rank-based agreement between predicted and observed enhancer activity values. Finally, the coefficient of determination (R^2) was used to estimate the proportion of variance explained by each model.

In addition to overall performance evaluation, the best-performing model was analyzed separately for each cell type to investigate differences in enhancer predictability across cellular contexts

4 Results

4.1 Comparison of model variants

The performance of all evaluated model variants is summarized in Table 1. Among the four architectures, the embedding-based model achieved the best overall performance across nearly all evaluation metrics. It obtained the highest Pearson correlation (0.489), Spearman correlation (0.480), and coefficient of determination ($R^2 = 0.239$), while also achieving the lowest MAE (0.577) and RMSE (0.769).

The full model, which combined cell-type embeddings with attention, achieved performance comparable to the embedding model, reaching a Pearson correlation of 0.486 and an R^2 value of 0.233. In contrast, the attention-based model performed worse than both embedding-containing architectures and did not improve upon the sequence-only baseline model.

Overall, models incorporating cell-type embeddings consistently outperformed models relying solely on sequence information, whereas the addition of attention mechanisms did not result in substantial performance gains.

Table 1: Performance of evaluated model variants on the test set.

Model	Pearson	Spearman	R^2	MAE	RMSE
Baseline	0.433	0.370	0.186	0.607	0.795
Embedding	**0.489**	**0.480**	**0.239**	**0.577**	**0.769**
Attention	0.408	0.362	0.166	0.608	0.805
Full	0.486	0.468	0.233	0.579	0.772

4.2 Training dynamics

Validation loss curves for all model variants are shown in Figure 1. All models exhibited a rapid decrease in validation loss during the first several epochs, followed by a slower improvement phase.

The embedding-based model achieved the lowest final validation loss and maintained the most consistent improvement throughout training. The full model showed a similar training trajectory, although its final validation loss remained slightly higher. In comparison, the baseline and attention-based architectures converged to noticeably higher loss values.

These results indicate that incorporating cell-type embeddings resulted in lower validation loss.

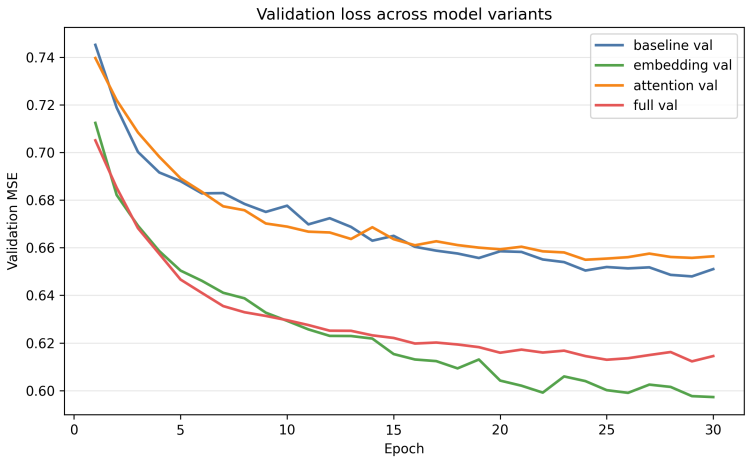

Figure 1: Validation loss across model variants during training.

4.3 Cell-type-specific performance

To investigate whether enhancer activity was equally predictable across all cellular contexts, the best-performing model was evaluated separately for each cell type. The resulting R^2 values are shown in Figure 2.

Performance differed substantially between cell types. WTC11 achieved the highest R^2 value ($R^2 = 0.271$), followed by K562 ($R^2 = 0.186$) and HepG2 ($R^2 = 0.113$), indicating that the model captured relative variation most effectively for WTC11.

These results indicate that enhancer activity was not equally predictable across the three cellular contexts and suggest that certain cell types may contain stronger sequence-associated regulatory signals than others.

4.4 Predicted versus observed enhancer activity

Figure 3 shows the relationship between predicted and observed enhancer activity values for the embedding-based model.

A positive association between predictions and measured activities was observed across the full range of values, consistent with the correlation metrics reported in Table 1. However, prediction accuracy decreased for highly active enhancers, with the model tending to underestimate extreme activity values. In contrast, predictions for low- and moderately-active enhancers were more tightly clustered around the expected trend.

Despite these limitations, the model captured a moderate proportion of the overall variation in regulatory activity across the test set.

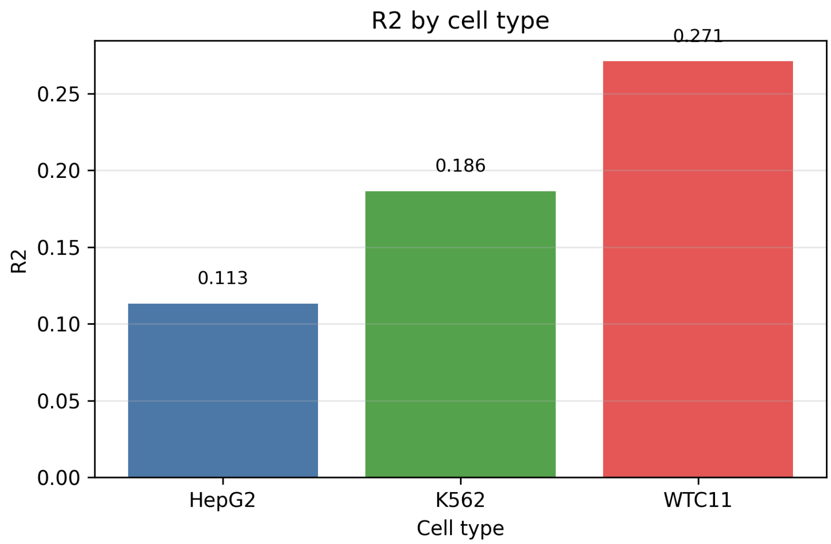

Figure 2: Coefficient of determination (R^2) obtained by the best-performing model for each cell type.

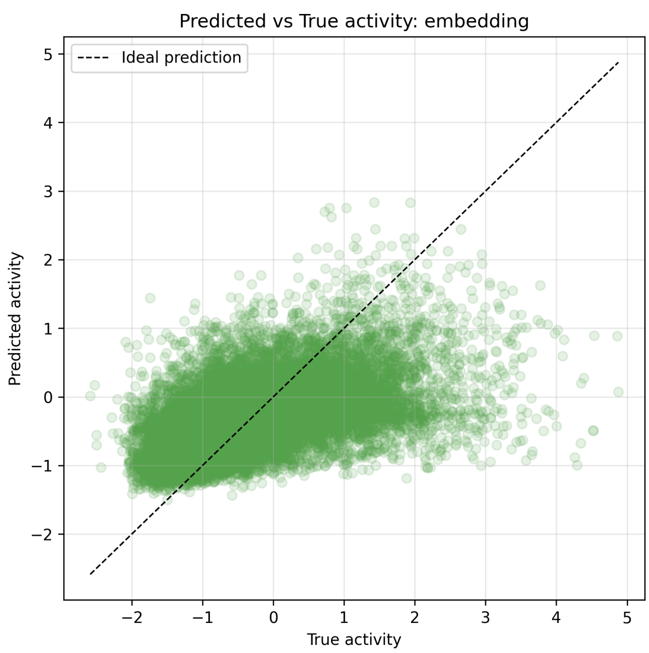

Figure 3: Predicted versus observed enhancer activity for the embedding-based model. The dashed line indicates perfect prediction.

5 Discussion

The primary objective of this study was to investigate whether explicit cell-type information improves enhancer activity prediction beyond what can be achieved using DNA sequence alone. The obtained results support this hypothesis. On the combined test set, models incorporating cell-type embeddings outperformed the sequence-only baseline across all reported evaluation metrics, achieving higher correlation coefficients and lower prediction errors. In contrast, attention-based architectures provided limited improvement and, when used alone, performed worse than the baseline model.

The superior performance of embedding-based models suggests that enhancer activity cannot be fully inferred from DNA sequence alone. While sequence features contain substantial information about regulatory potential, enhancer function is strongly influenced by cellular context. By incorporating learned cell-type embeddings, the model was able to utilize additional information about the biological environment in which a given enhancer was assayed. This indicates that explicit representation of cellular context contributes meaningful predictive information beyond sequence-derived features.

Interestingly, the attention-based model did not improve prediction performance despite increasing architectural complexity. One possible explanation is that the convolutional layers were already sufficient to capture the most informative sequence motifs and local regulatory patterns present within the enhancer sequences. Under these conditions, the attention mechanism may not have provided additional useful information for the prediction task. Furthermore, the relatively short sequence length used in this study may have limited the potential benefits of attention-based pooling.

Performance also varied considerably across cell types. WTC11 achieved the highest Pearson correlation and R2, followed by K562 and HepG2, indicating that relative variation was captured most effectively for WTC11. These differences suggest that enhancer activity may not be equally predictable across cellular contexts. One possible explanation is that regulatory activity in WTC11 cells is more strongly associated with sequence-encoded features, making enhancer activity easier to predict from DNA sequence and cell-type information. Alternatively, differences in experimental variability or regulatory complexity between cell types may have contributed to the observed performance differences. Another possible explanation for the higher predictive performance observed in WTC11 cells could be the biological origin of the cell line. WTC11 cells are induced pluripotent stem cells derived from a healthy donor, whereas HepG2 and K562 are cancer-derived cell lines. The regulatory programs of cancer cells are often altered by mutations, chromosomal abnormalities, and dysregulated transcriptional networks, potentially making enhancer activity more difficult to predict from sequence information alone. However, additional analyses would be required to determine whether these biological differences are directly responsible for the observed variation in predictive performance.

Despite the improvements obtained through cell-type embeddings, the overall predictive performance remained moderate, with the best model achieving an R^2 value of 0.239. Scatter plot analysis further revealed a tendency to underestimate highly active enhancers. These observations indicate that a substantial fraction of enhancer activity remains unexplained by the information available to the model. This is not unexpected, as enhancer function depends on numerous biological factors that were not included in the present study, including chromatin accessibility, transcription factor abundance, epigenetic modifications, and three-dimensional genome organization.

Several directions could be explored in future work. Additional genomic and epigenomic features could be incorporated alongside DNA sequence to provide a more comprehensive representation of cellular context. More advanced architectures, such as transformer-based models, could also be investigated. Finally, expanding the analysis to a larger number of cell types may provide further insight into the relationship between cellular context and enhancer activity prediction.

6 Conclusion

This study evaluated four convolutional neural network architectures for predicting enhancer activity from DNA sequence data obtained from MPRA experiments across three human cell types. The results demonstrated that incorporating explicit cell-type information through learned embeddings improved aggregate predictive performance on the combined test set relative to the sequence-only model. In contrast, attention mechanisms provided limited benefit and did not improve prediction accuracy when used independently.

The best-performing model achieved the highest correlation coefficients and the highest coefficient of determination, indicating that cellular context contains predictive information that cannot be fully inferred from sequence alone. Additionally, prediction performance differed across cell types, with WTC11 achieving the highest Pearson correlation and R2.

Overall, the findings support the importance of incorporating cell-type information into computational models of gene regulation and demonstrate that even simple representations of cellular context can improve enhancer activity prediction.

7 AI Assistance Statement

Generative AI tools were consulted on an ad hoc basis when clarification was needed or when difficulties were encountered during project development. Their use was limited to explaining programming concepts and suggesting possible approaches to implementation problems. All methodological decisions, code implementation, analyses, and interpretation of results were performed and verified by the author.

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